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CMi EPFL Center of MicroNanoTechnology

☑ Semestral Project ☐ Master Project ☐ Thesis ☐ Other

# Microfluidics for Si-Nano Wire Reconfigurable FET biosensor

## Description of the fabrication project

In order to further automate the functionalization, washing, and detecting steps, we need a PDMS microfluidic layer to both cover the nearby electrodes and guide the solution to go through the active region of the biosensor. Initially, microfabrication at wafer level will be done and then, with a more optimized design, fabrication on chip will be preferred for to better match device topography.

| **Technologies used**<br>*!! remove non-used !!* | | | |
|---|---|---|---|
| SU-8 photolithography at wafer level, PDMS line, O2 activation for bonding. | | | |
| **Ebeam litho data - Photolitho masks - Laser direct write data (.gds file in attachment)** | | | |
| **Mask #** | **Critical Dimension** | **Critical Alignment** | **Remarks** |
| M1 | **2um** | <1um | PhotoL: SU-8 patterning for PDMS mold |
| **Substrate Type** | | | |
| Silicon test wafer, Si/100mm/SSP/Test<br>Sacrificial dummy chips (10x10mm, later) | | | |

## Interconnections and packaging of final device

Thinning/grinding/polishing of the samples is required at some stage of the process.

☑ No ☐ Yes => confirm involved materials with CMi staff

Dicing of the samples is required at some stage of the process.

☐ No ☑ Yes => confirm dicing layout with CMi staff

Wire-bonding of dies, with glob-top protection, is required at the end of the process.

☑ No ☐ Yes => confirm pads design (size, pitch) and involved materials with CMi staff

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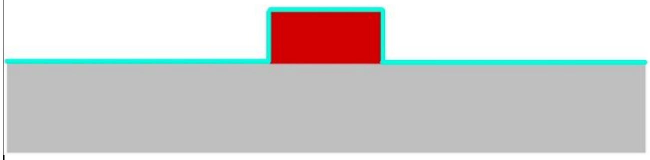
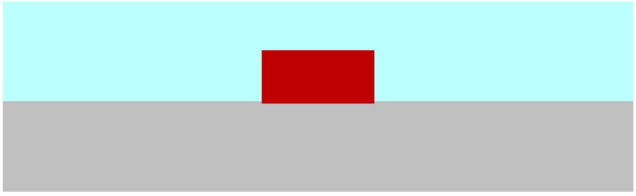
## Step-by-step process outline

| Step | Process description | Cross-section after process |
|---|---|---|
| 1 | **Description:** Wafer preparation for SU-8 photolithography, O2 plasma (cleaning and activation).<br>**Machine:** Z11/ Tepla 300 |  |
| 2 | **Description:** SU8 coating + soft bake<br>**Material:** SU8 KAYAKU<br>**Machine:** Z13/Sawatec SM 200 (Manual Spin Coating) + Z13/Sawatec HP200 (Soft Bake)<br>**Thickness:** 30um. |  |
| 3 | **Description:** Direct laser exposure<br>**Material :** SU8.<br>**Machine:** Z16/MLA150-1.<br>**Remark:** Need to perform dose testing. |  |
| 4 | **Description:** PEB + Development<br>**Material:** SU8, PGMEA<br>**Machine:** Z13/Sawatec HP200 (PEB) + Z13/Wet Bench |  |
| **Remark:** For later stages, molds can also be fabricated using 3d printing with clear photoresist or with LF55GN photolithography in DLL labs at MED buildings. | | |

| | | |
|---|---|---|
| **5** | **Description:** TMCS passivation **Machine:** PDMS line(Z12): TMCS desiccator. |  |
| **6** | **Description:** PDMS mixing, degassing, pouring and curing **Material:** 5mm of PDMS **Machine:** Z12/PDMS dispenser, Thinky mixer, Desiccator, 6800 spin coater, Oven. |  |
| **7** | **Description:** Demolding and hole punching **Machine:** PDMS line(Z12), Cutting tools. |  |
| **8** | **Description:** PDMS alignment, O2 activation and bonding **Machine:** Z12/O2 Plasma, glass slide, oven. |  |